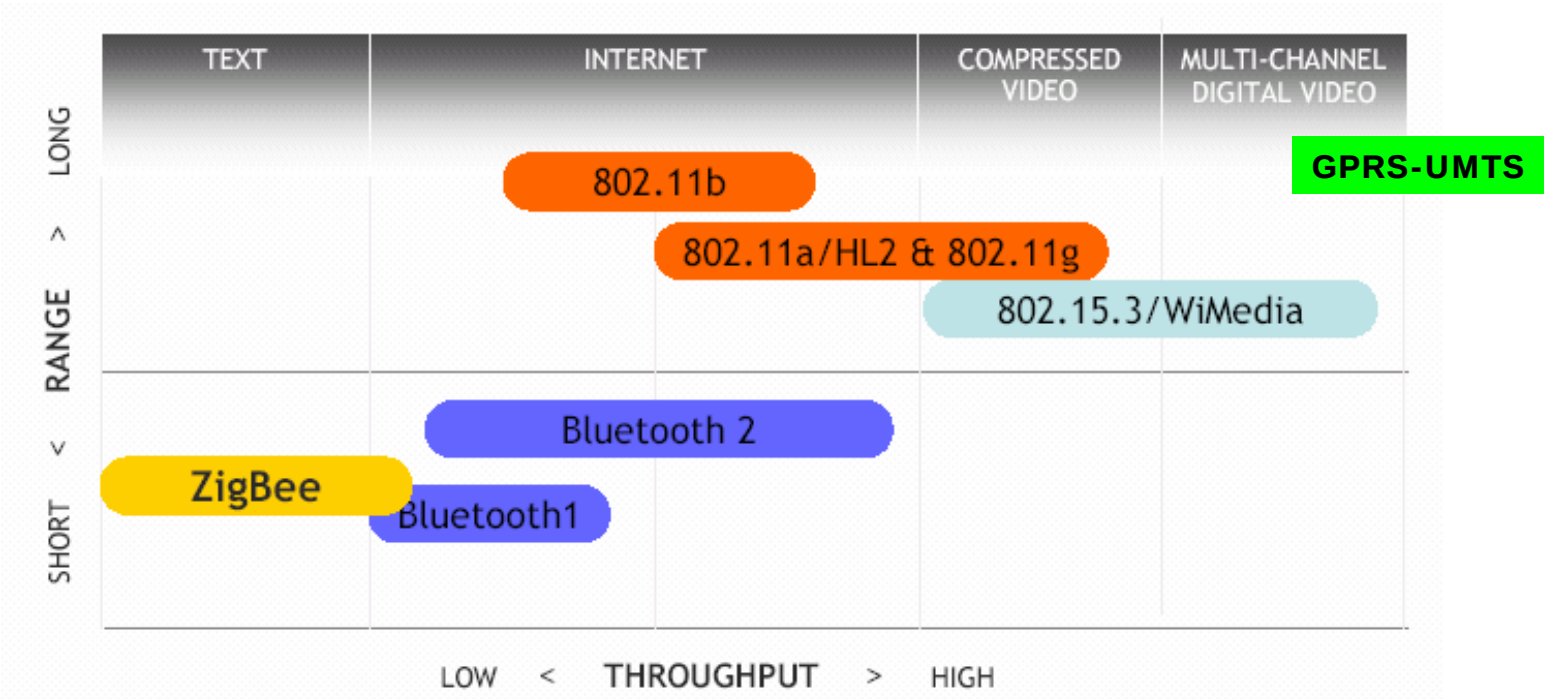
Wireless Technologies

Outline

- GPRS-UMTS
- WiFi (802.11)
- Bluetooth
- ZigBee



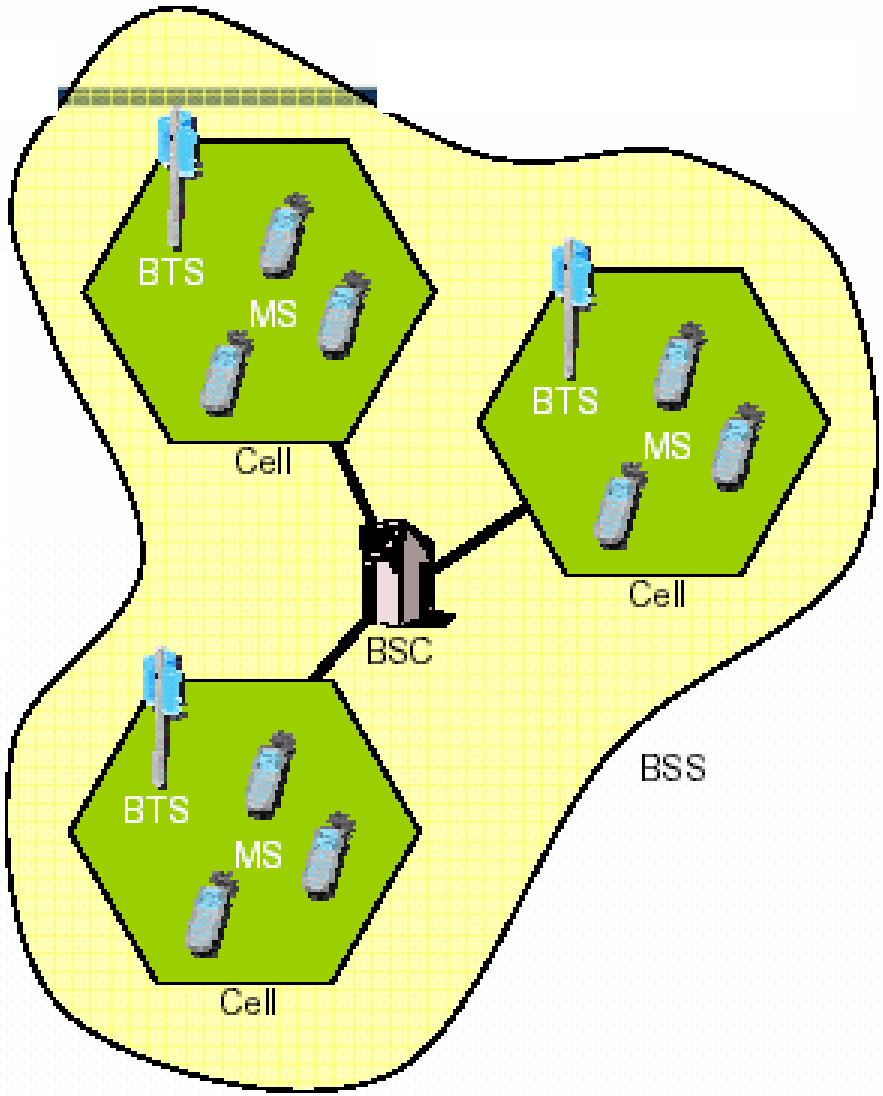
GSM, GPRS, UMTS

- Mobile phone networks are nearly ubiquitous
- Still basic client-server architecture
  - Only last-mile wireless
  - Wired backbone structure
- From Voice to data
- Cellular network evolution
- 1G networks (analog cellular networks)
- 2G networks (GSM simple data services)
- 2.5G networks (GSM with GPRS)
- 3G networks (UMTS)

2G – GSM (1982)

- GSM, Global System for Mobile communication
- Goals
  - Good speech quality
  - Low cost for terminals and services
  - International roaming
  - Handheld terminals
- Primary usage
  - Voice call + SMS
- Network
  - Circuit switched network
  - Three frequency bands (900, 1800, 1900 MHz)

Cellular Network

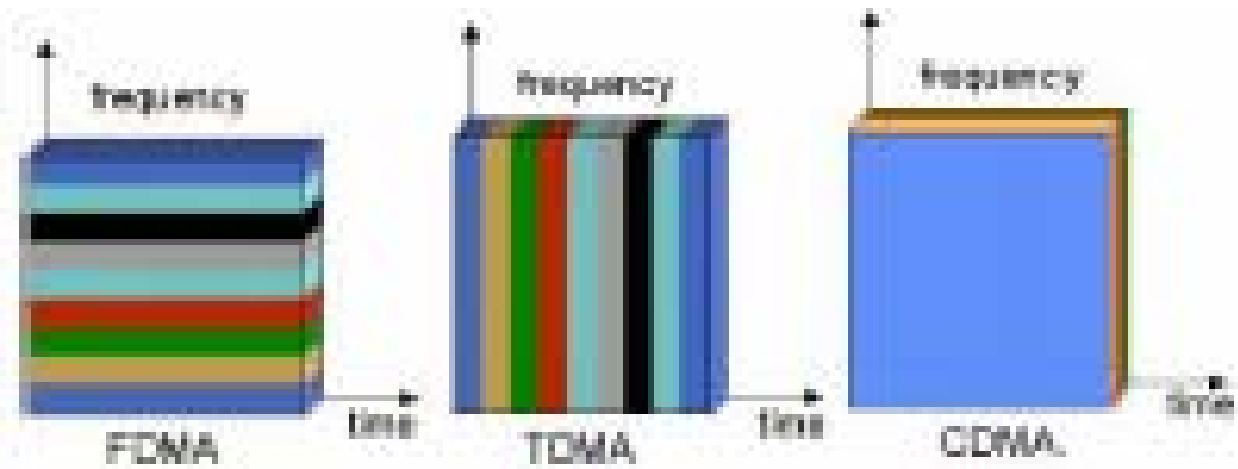


GPRS (General Packets Radio Services)

- GPRS implements packet switched data
  - Overlay on existing GSM network
  - Always ON
  - Supports bursty applications such as email, web
- Applications
  - Communication (internet, mail, messaging)
  - Information services
  - Localization-based services

UMTS (Universal Mobile Telecommunication System)

- 3G mobile network
- Supports voice and data services
  - 144 kbps – rural
  - 384 kbps – urban
- QoS support
- Spread-spectrum modulation

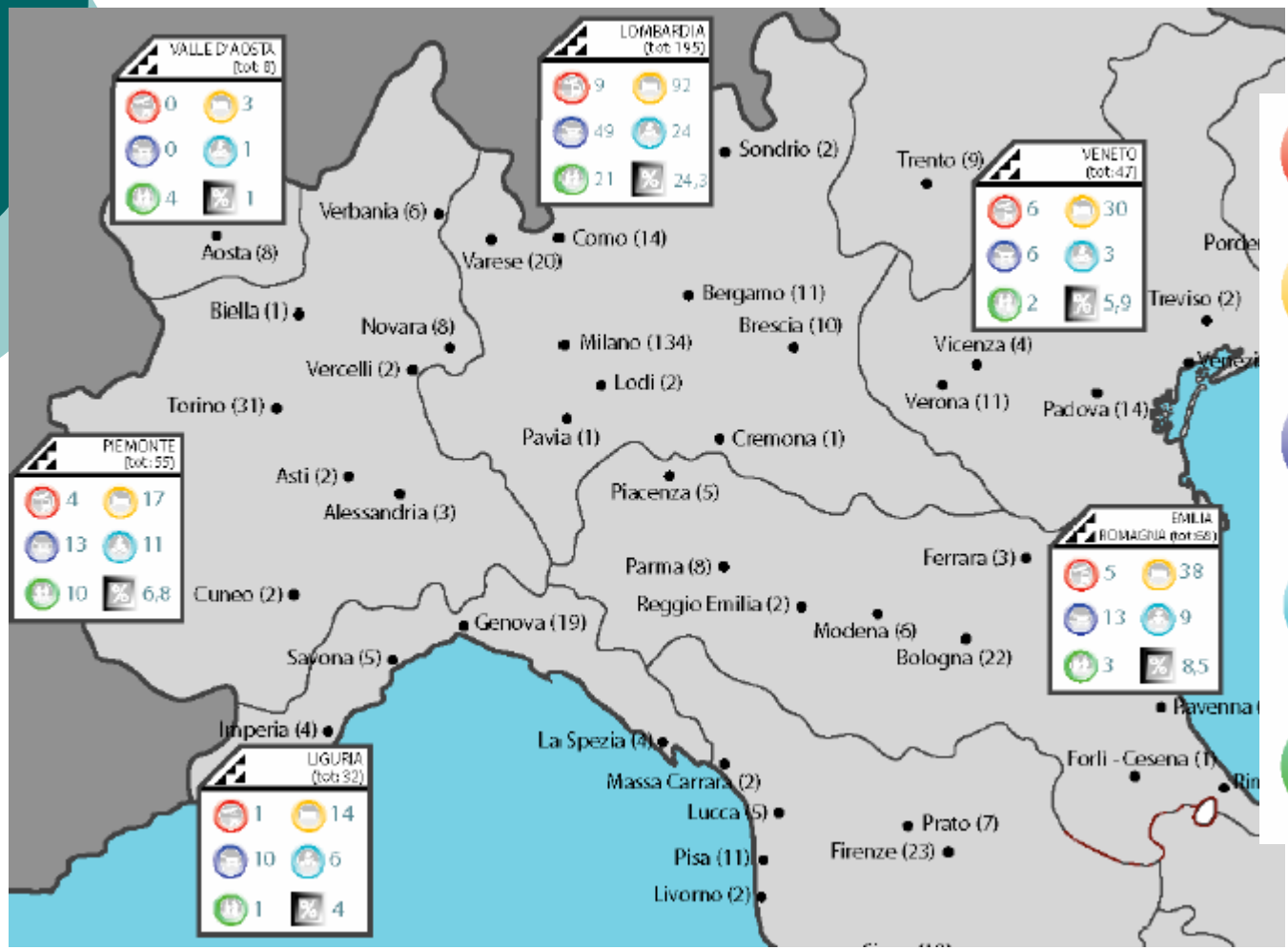


WiFi

- Most rapid growth of all wireless technologies
- Expected market growth
  - $ 1.79 billion (2001)
  - $ 3.85 billion (2004)
- More than 70% all notebooks with 802.11 already embedded (2004)
- Wireless hot-spots begin to pervade



Hot-spot in Italy



aeroporti e stazioni ferroviarie e porti (tot: 39)

hotel e centri congressi (tot: 375)

retail commerciali (tot: 160)

pubblici esercizi (tot: 95)

altro (tot: 134)

Applications

- Primary application domain is traditional networking at home and in the office
- Wireless video connections between set-top boxes and TV sets
- Video streaming from camcorder/camera to TV set and PC
- Exchange and transmission of radio (network radio)
- Wide-area mobile networks
- Others?

802.11 Extensions

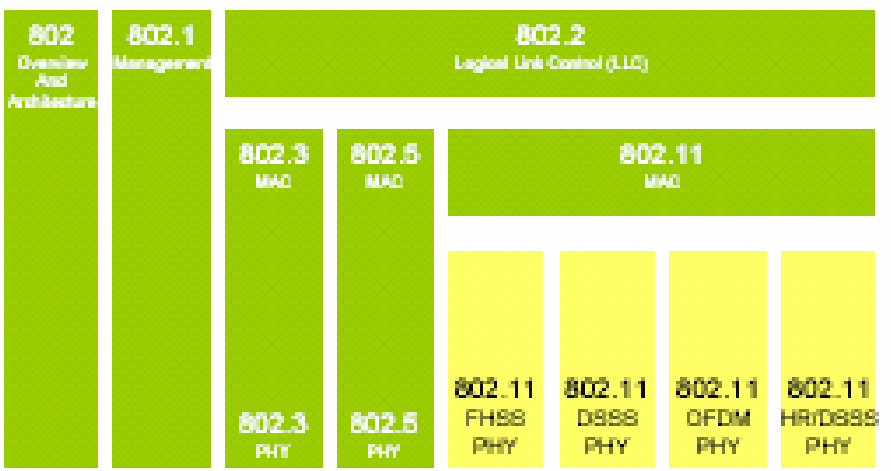
- **802.11** original standard provided for 1- and 2-Mbps PHY layer, CSMA/CA (1997)
- **802.11a** Enhancement to provide 54 Mbps in the 5 GHz band (1999)
- **802.11b** Enhancement to provide 11 Mbps in the 2.4 GHz band (1999)
- **802.11d** Changes for international regulatory compliance (2001)
- **802.11e** Enhancements to the MAC layer to provide QoS through prioritized CSMA and advanced polling techniques
- **802.11f** Recommended practices for inter-access point communication
- **802.11g** PHY layer enhancement to provide 54 Mbps in the 2.4 GHz band
- **802.11h** Enhancement to 802.11a to achieve regulatory compliance with Europe
- **802.11i** Security enhancement
- **802.11j** Changes to meet Japanese regulatory requirements
- **802.11k** Improved WLAN system management
- **802.11m**
  A generalized cleanup and editing of the existing standard
- **802.11?**
  More than 100 Mbps throughput

Throughput and Capacity

WLAN Mode	Maximum Link Rate	Maximum UDP Rate	Maximum TCP Rate	System Capacity	
				# of Channels	Maximum UDP Capacity
802.11a Turbo	108 Mbps	55.1 Mbps	42.7 Mbps	6	330.6 Mbps
802.11a	54 Mbps	30.7 Mbps	24.0 Mbps	13	399.1 Mbps
802.11g (11g-only)	54 Mbps	30.7 Mbps	24.0 Mbps	3	92.1 Mbps
802.11g (with 11b present and idle)	54 Mbps	19.6 Mbps	14.5 Mbps	3	58.8 Mbps
802.11g (with 11b present and active)	54 Mbps	11.2 Mbps	9.2 Mbps	3	33.6 Mbps
802.11b	11 Mbps	7.1 Mbps	6.1 Mbps	3	21.3 Mbps

McFarland and Wong

Relation to the OSI model



Applications

7. Application Layer

6. Presentation Layer

5. Session Layer

4. Transport Layer

3. Network Layer

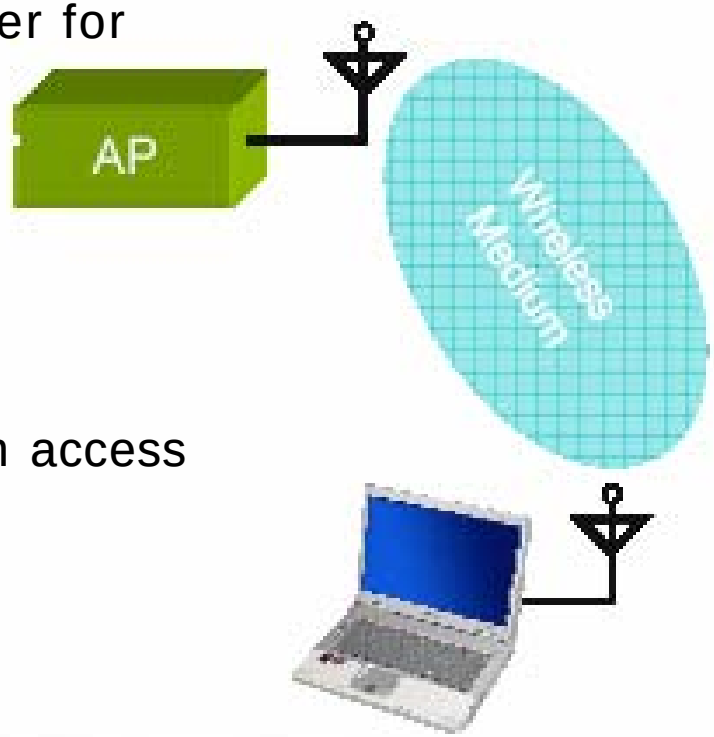
2. Link Layer

1. Physical Layer

Communication Network

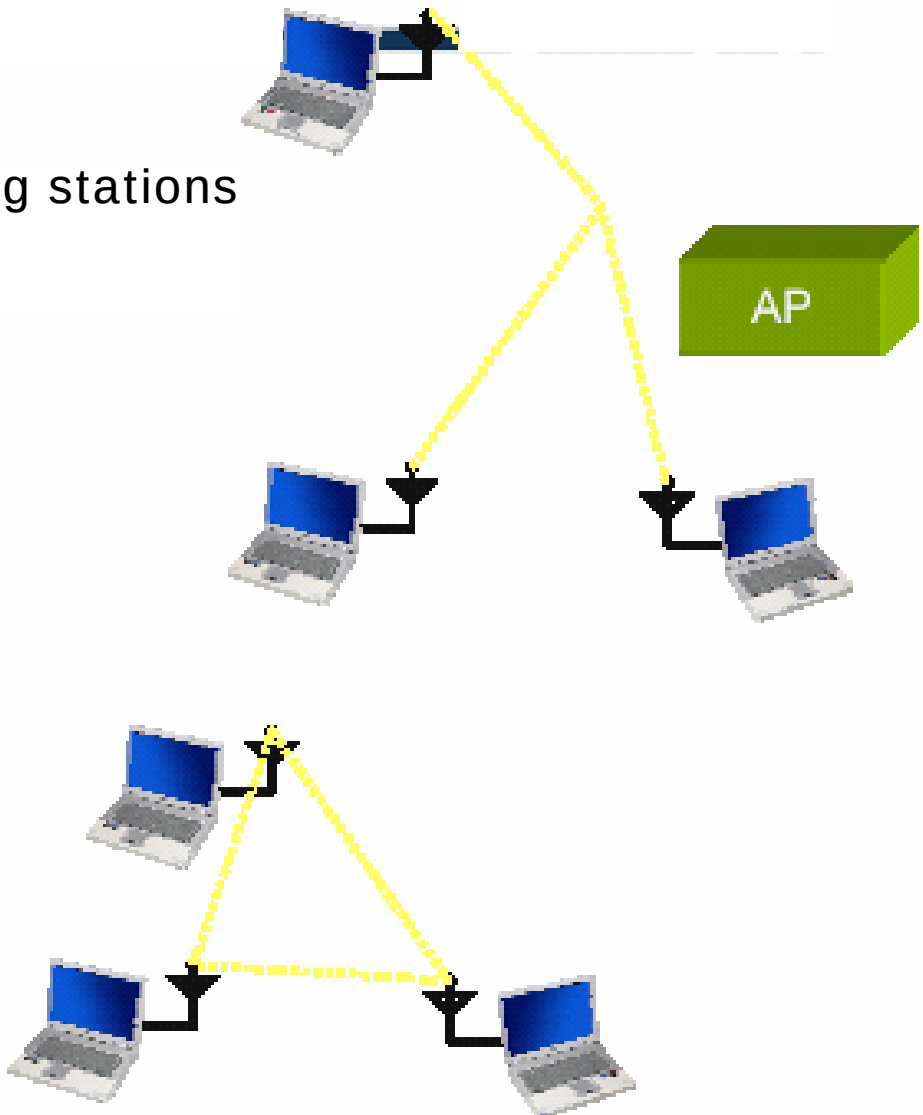
Major Components

- Station
  - Computing device with wireless network interface
- Wireless medium
  - Radio frequency (RF) physical layer for communication
- Access Point (AP)
  - Wireless-to-wireless bridging
  - Wireless-to-wired bridging
- Distribution System
  - Forwarding frames (packets) from access point to access point



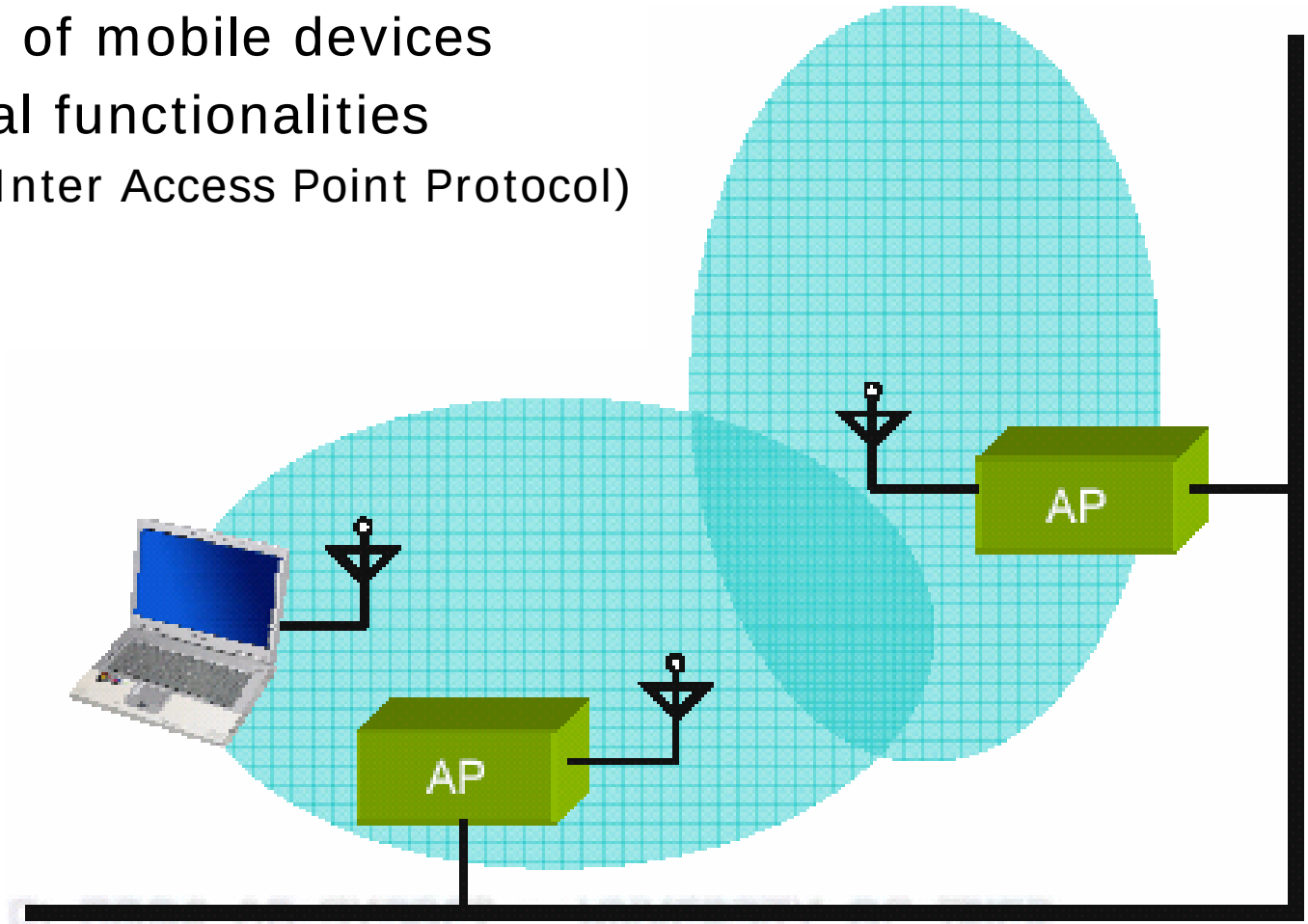
Network types

- Basic service set (BSS)
  - A group of communicating stations
- With access point
  - Infrastructure Network
  - Infrastructure BSS
- Without access point
  - Independent network
  - Independent BSS (IBSS)
  - Ad-hoc network



Extended Service Sets

- Bridging between APs
- Roaming of mobile devices
- Additional functionalities
  - IAPP (Inter Access Point Protocol)



Problems in wireless communication

- Available unlicensed spectrum allocation (government regulation)
  - Unlicensed ISM bands (Industry Scientific Medicine)
- Only low transmission power levels allowed
  - Spread spectrum modulation
- Multi-path echoes
- Interference
- Noise

WiFi Additional Features

- Power saving mode
  - Mobile station may enter power saving mode
  - Access point will buffer frame for sleeping devices
  - Upon wakeup AP will send buffered frames
- WEP (Wired equivalent privacy)
  - Based on RCA
  - 64bit keys
  - Already cracked!!

Bluetooth

- The term Bluetooth refers to an open specification for a technology to enable short-range wireless voice and data communications anywhere in the world.
- Early 1998 – Special Interest Group (SIG) formed
  - Code name: Bluetooth
    - Harald Blatand “Bluetooth”, Viking King of Denmark (940-981 aC) that unified European northern countries
  - Promoter companies: Ericsson, IBM, Intel, Nokia and Toshiba
- 1999 – Bluetooth 1.0 Specification Release
  - 10 – 100 meters networking
- Today – Bluetooth 2.0 work is ongoing
  - Promoter Companies: 3Com, Ericcson, IBM, Intel, Lucent, Microsoft, Motorola, Nokia, Toshiba.
  - And other 1883 companies in the SIG

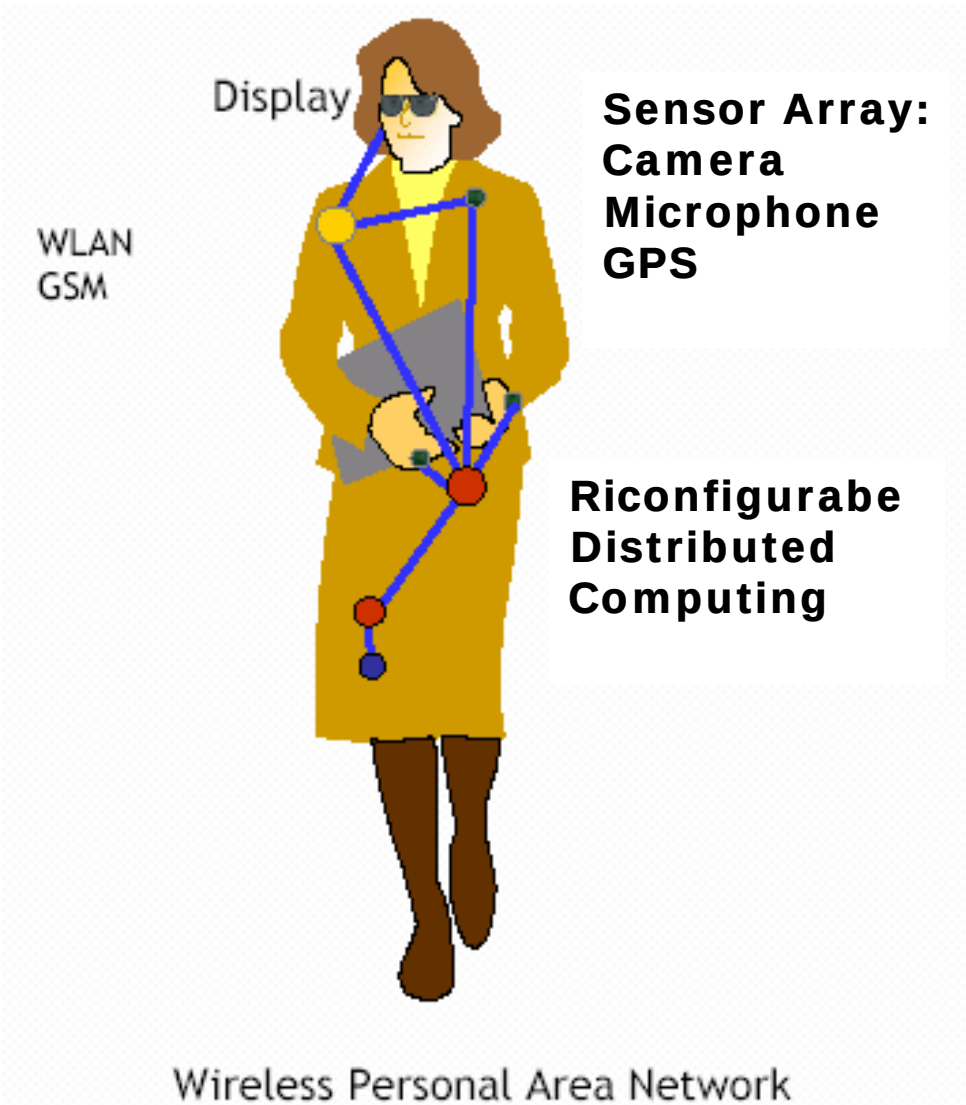


Bluetooth Usage Models

- Bluetooth as a cable replacement
  - The cordless computer
  - The instant postcard
- Supporting voice communication
  - Ultimate headset
  - Three-in-one phone
  - The speaking laptop
- Networking
  - The interactive conference (file transfer)
  - The internet bridge (dial-up, direct access)
  - The automatic Synchronizer
  - Ad-hoc networking
- Hidden Computing

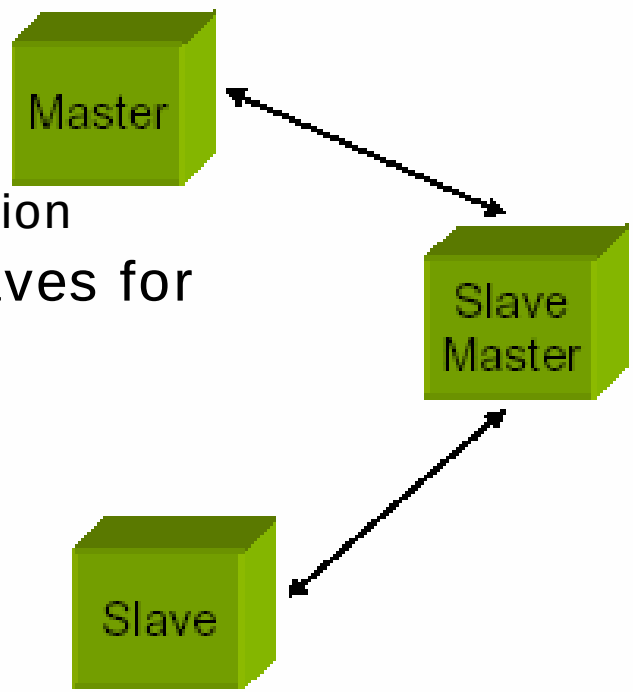


Personal Area Network



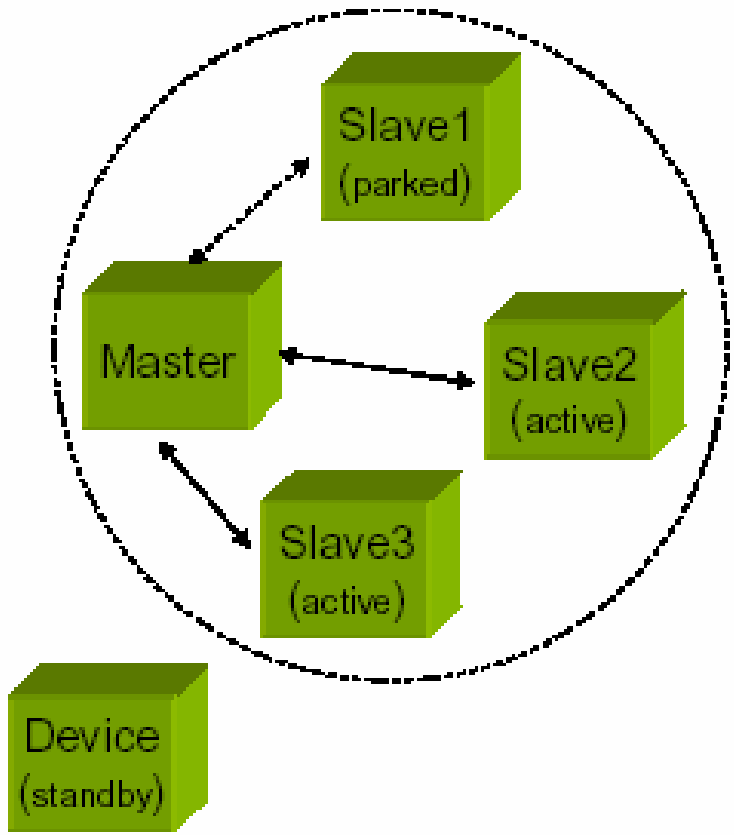
Master and Slave Role

- For each link one master and one slave
- Role of the master
  - Determines frequency hopping (spread spectrum modulation) based on its address
  - Polling slaves to transmit and receive
  - No special privilege
- Role of the slave
  - Follow the hop sequence
  - Responsiveness VS. power consumption
- Devices may act as master and slaves for different links
- Parked nodes



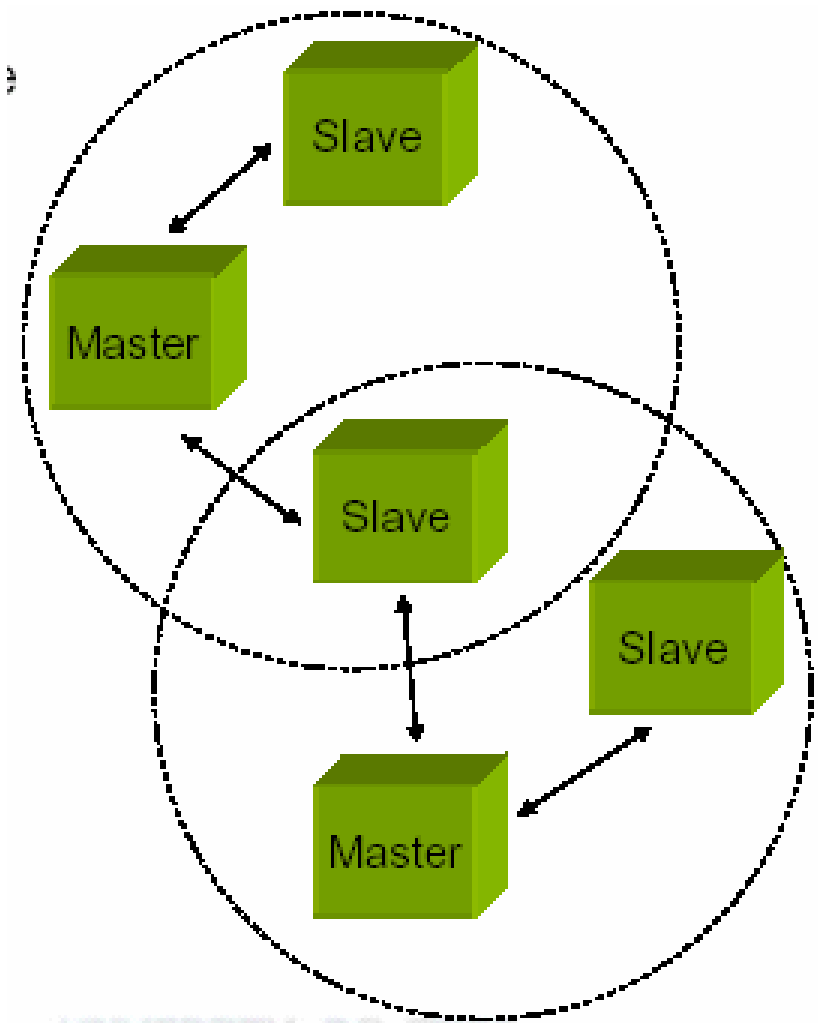
Piconet

- Master may communicate with multiple slaves
  - 7 active, 255 parked
- Piconet is defined by one master and its slaves
  - All slaves follow the same hopping sequence
  - Not all devices in the proximity of the master are in the Piconet
- Typical Piconets are expected to have few devices



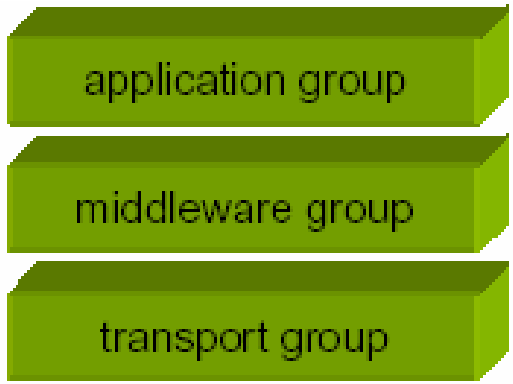
Scatternet

- Device mat take part in more than one Piconet
- Scatternets are defined by such overlapping piconets
  - Piconets remain as described
  - Each Piconet has its own hopping sequence
- Different roles and states are possible



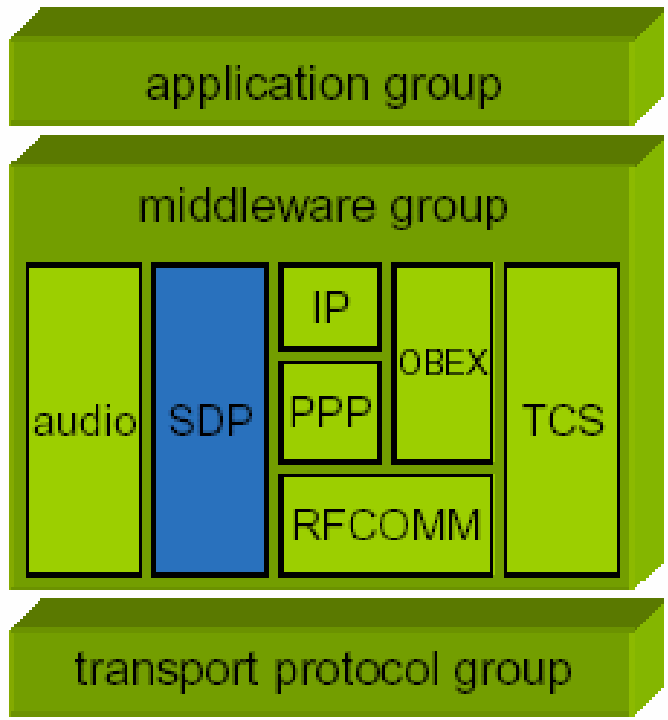
Protocol Stack Components

- Transport protocol group
  - Locate other devices
  - Create, configure and manage both physical and logical links
  - Transport of data from higher-layer protocols and applications (asynchronous/synchronous)
- Middleware protocol group
  - Provide existing and new applications
  - Existing protocols like PPP, IP, TCP, OBEX, ...
  - New Bluetooth aware protocols like RFCOMM, TCS (voice), SDP (service discovery)
- Application group
  - Legacy applications unaware of Bluetooth (e.g. modem dialer, web browser)
  - Bluetooth aware applications (e.g. telephony control via TCS)
  - Bluetooth profiles (advanced API): file transfer,...



Service Discovery Protocols

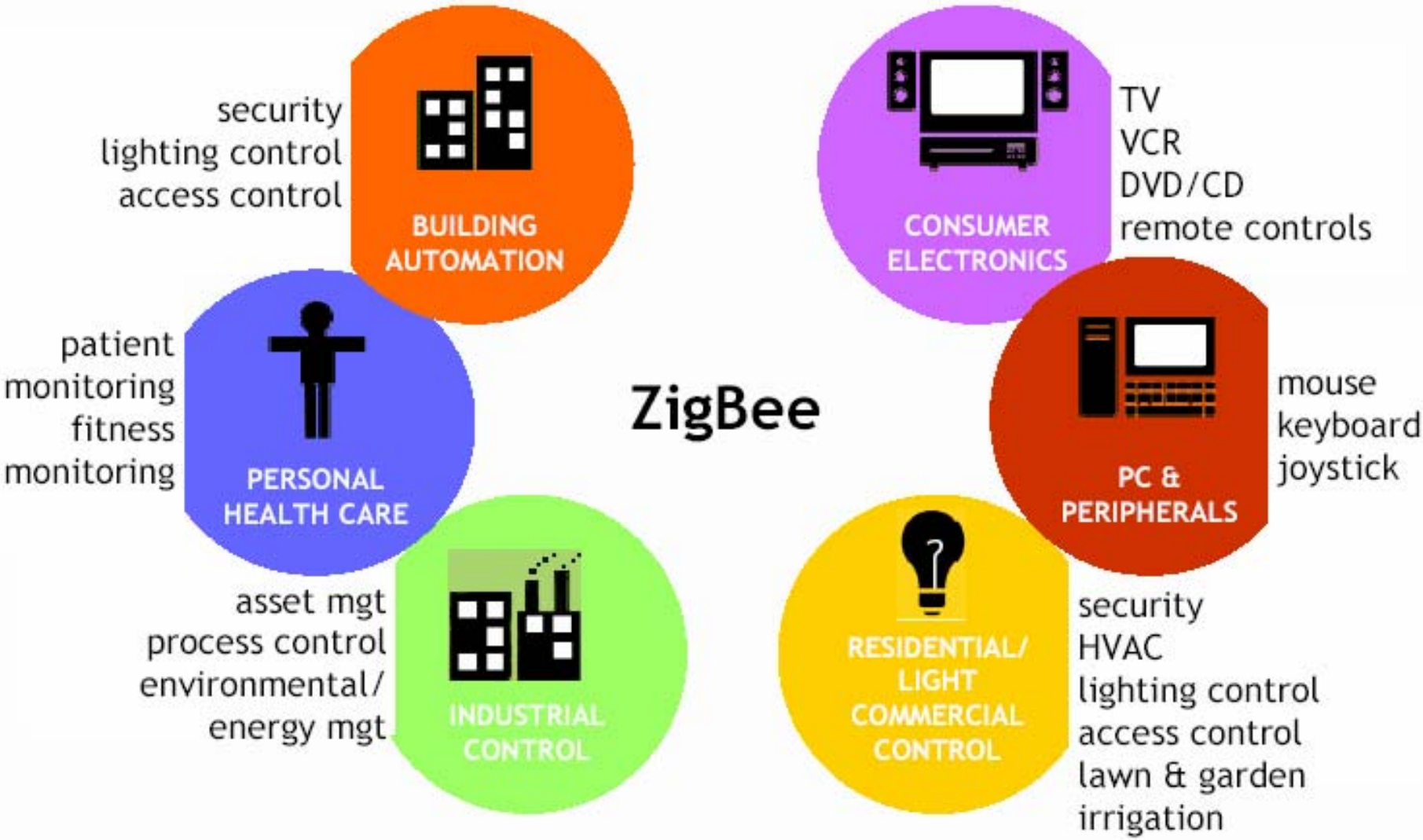
- Services in traditional networks: file servers, printer servers, naming
- Static configuration insufficient for dynamic ad-hoc networks
- SDP provides standard methods to discover services at connected devices
- Symmetrically, SDP enables description of own services



Zigbee

- Low-power communication
  - 10-100 meters wireless tunable range
- Physical Layer MAC from IEEE 802.15.4
- Zigbee SIG upper levels
- Zigbee open standard
- Mesh networks

Zigbee Application Scenarios



Mobile Ad-Hoc Networks

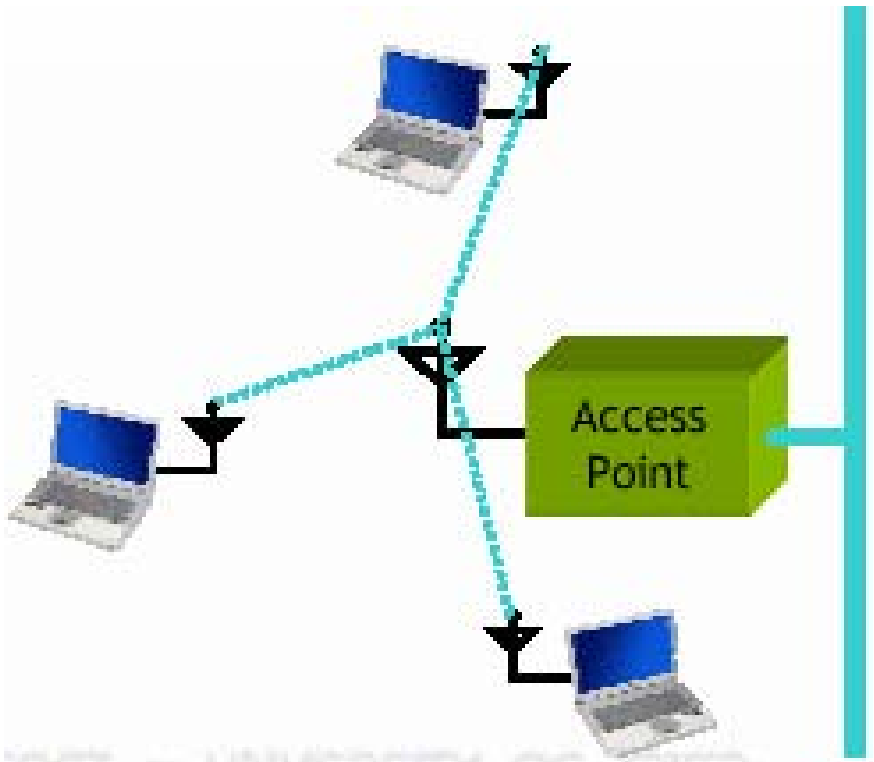


Characteristics

- Small devices
  - Mobile
  - Autonomous
  - Own, but limited power supply
  - Wireless Communication
- Spontaneous networking
  - Traditional administration impossible
  - One-time acquaintances are common
- No wired infrastructure scientifically challenging
- In reality hybrid systems with access to wired networks

Single-hop networks

- State of the Art
  - Example: WLAN
  - Infrastructure BSS and Extended SS
  - GSM and UMTS
- Wireless last mile
  - TCP/IP, Mobile IP, ...
  - Client/Server, WWW, ...
  - Web Services
- QoS for
  - IP Telephony
  - Audio and Video



Applications…

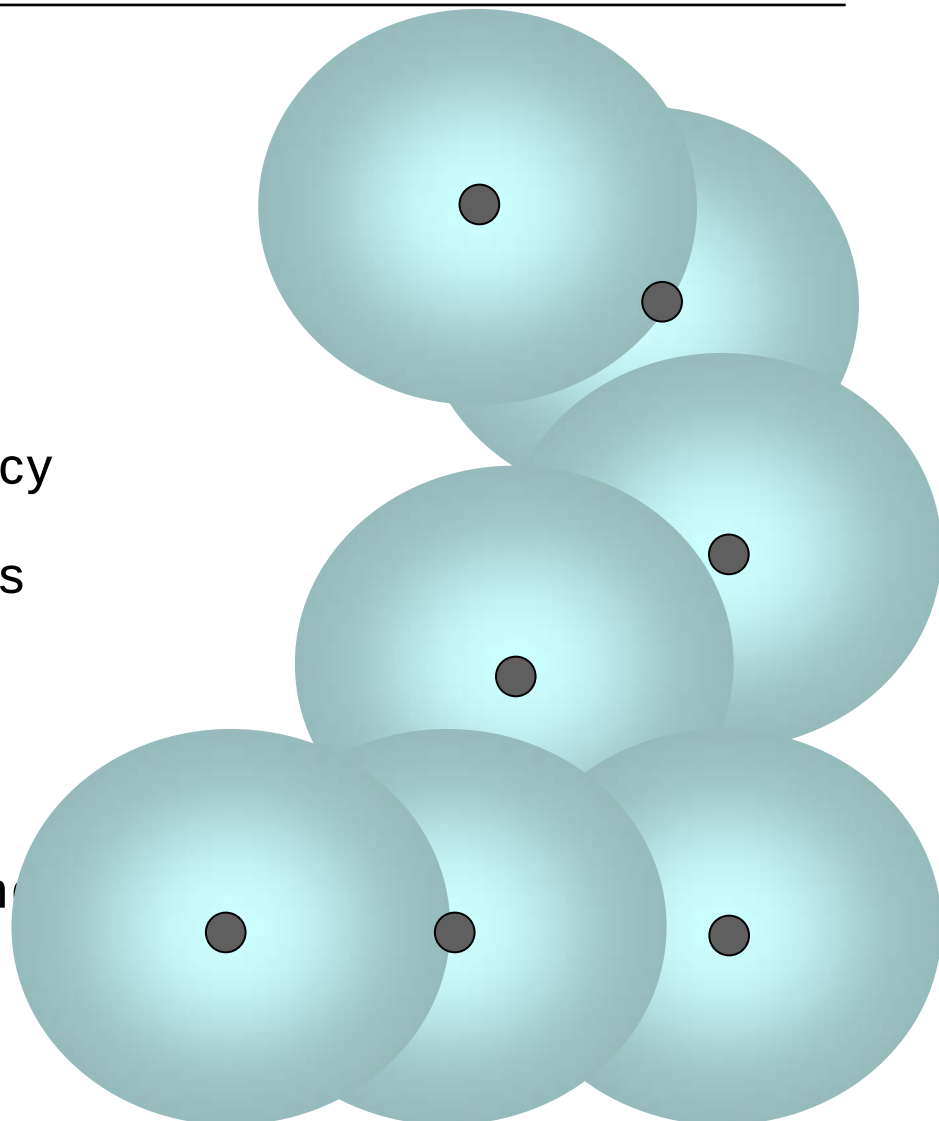
- Wireless access at the university campus
- Delivering e-learning and online lectures
- File-sharing
- …





Multi-hop Networks…

- Unreliable communication
  - Frequent topology changes
  - High Error rate (Multi-path, Interference, …)
- Natural structure for 1010 nodes and more
  - No wired infrastructure
- Essential in case of emergency and disaster
- New programming paradigms required
  - Self-organization, Autonomy
  - Altruism, Synergy
  - Context Awareness
  - Power Awareness
- Requires sophisticated routin



A Vision of the Future…

- Consider a scenario a few years hence in which a large city like Boston might have several wireless base stations in every building - a number of nodes in the order of 10^7. If most of the electrical devices in the buildings and those carried on by people are wirelessly networked too, then the total number of nodes could be as high as 10^10. If these nodes communicate peer-to-peer with nearby devices, then one could envision the entire city as connected into a mobile ad-hoc network approximately 10^3 hops in diameter.

Applications

- File-Sharing
- Profile Sharing (Toothing)
- Familiar Stranger (Intel Berkeley)
  - Maintain a history of the encountered devices
  - Compare current situation with history

Fleet Net

- German Research Project
  - Inter-Vehicle Communication System
  - MANET
- Service and Applications
  - Exchange sensor data between vehicles (e.g. road condition)
  - Emergency breaking
  - Coordinated lane entering
  - Traffic Jams
- Candidate Wireless Technologies
  - WiFi
  - UMTS

Fleet Net Scenario

